

Lucas Lu

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EDUCATION

Southern University of Science and Technology (SUSTech) Aug. 2025 – Present
B.M. Candidate in Industrial Engineering Shenzhen, China

- **GPA: 3.91 / 4.0 | Major Ranking: 1 / 11 | Comprehensive Assessment Ranking: 3 / 244.**
- **Selected Coursework:** Mathematical Analysis I–II (93, 96), Ordinary Differential Equations B (98), Foundation of Probability Theory (94), Introduction to C Programming (96), Introduction to Machine Learning and Big Data Analytics(P).

RESEARCH INTERESTS

Operations Research & Optimization; Emergency Response & Logistics.

PROJECT EXPERIENCE

EMS Station Location Optimization 2026 – Present
Advised by Prof. Rowan Wang Shenzhen, China

- Optimized EMS station locations across Shenzhen; partitioned the city into 5,279 500m×500m grids.
- Built an XGBoost ETA model, predicting ambulance travel times with 90% accuracy.
- Constructed a detailed analysis of the EMS station cancellation and scheduling problem.

AED Deployment at Gulongzhong based on robust optimization 2026 – Present
Advised by Prof. Rowan Wang Shenzhen, China

- Compared fixed AED placement with a dynamic human–vehicle–drone response system using road-network analysis.
- Formulated a velocity distributionally robust optimization model to determine optimal locations
- First prize, China University Mechanical Engineering Innovation & Creativity Competition (2026)

Meal Optimization WeChat Mini Program 2026 – Present
Advised by Prof. Rowan Wang Shenzhen, China

- Built a personalized meal recommendation mini program based on a mixed-integer programming model.
- Added constraints like nutrition limits, disliked-food exclusions, dish swaps, and fallback plans.
- Improved reliability with carb alerts, solver compatibility, and clearer feedback.

LEADERSHIP & VOLUNTEER EXPERIENCE

Leadership

Head of Publicity, SUSTech Moviesalon | Social Media and news management Sep. 2026 – Present
Organizer, SUSTech IE Hunt | Optimization challenge planning and team coordination Oct. 2025 – Present
Class President, Industrial Engineering | Class management and daily affairs Sep. 2026 – Present
Sub-team Leader, Orange Light Volunteer Service Team | Community research and volunteering Mar. 2026 – Present
Student Representative, Zhicheng College 10th Anniversary Ceremony | Delivered a speech. Oct. 2025

Volunteer Experience

Asia-Pacific Economic Cooperation (APEC) Shenzhen 2026 June 2026 – Present
Association of Pacific Rim Universities Undergraduate Leaders Program 2026 Aug. 2026
Shenzhen Nan Shan Half Marathon (NSHM) 2025 Dec. 2025
Completed 160+ hours of volunteer service. Sep. 2025 – Present

SELECTED HONORS & AWARDS

- **First-Class Outstanding Student Scholarship**, SUSTech, top 5% university-wide Oct. 2026
- **First-Prize**, The 9th Industrial Engineering & Lean Management Innovation Competition Oct. 2026
Sub-competition of China University Mechanical Engineering Innovation & Creativity Competition 2026
- **First Prize**, GBA(The Greater Bay Area) Industrial Engineering Innovation Competition, top 2 of 54 teams Sep. 2026
- **Second Prize**, 17th National College Student Mathematics Competition, top 15% province-wide Dec. 2025
- **APRU ULP Outstanding Student Ambassador** (one of ten university-wide) June 2026
- **Outstanding Individual**, SUSTech Winter Social Practice, one of 50 selected university-wide Mar. 2026
- **Outstanding Individual**, SUSTech Winter Visit to Alma Mater Program Sep. 2026

SKILLS & LANGUAGES

- **Programming Skills:** Python, MATLAB, R, C, LaTeX, AnyLogic.
- **Languages:** Mandarin (Native), English (Proficient), Korean (Daily basics).